

SPECIFIC AIMS

Associative learning is a biologically adaptive process that allows organisms to learn that stimuli from the environment can predict positive outcomes like food or negative outcomes like the presence of a predator. This can occur via first- or higher-order conditioning. In first-order conditioning, conditioned responding develops to a cue that has been directly paired with reinforcement. In higher-order conditioning, cues gain the ability to elicit conditioned responses, not because they have been directly paired with reinforcement, but because they have been paired with other cues which have. It is important to understand the mechanisms underlying higher-order conditioning as it may be involved in maladaptive or destructive behavior such as overeating or drug seeking behavior.

The proposed work examines sensory preconditioning, a form of higher-order conditioning. In sensory preconditioning, Cue 1 and Cue 2 are first paired together in the preconditioning phase. After which, Cue 1 is directly paired with an unconditioned stimulus (i.e., first-order conditioning phase). With this procedure, Cue 2 can ultimately elicit a conditioned response at retrieval even though it was never paired with reinforcement. Recent work has implicated the retrosplenial cortex (RSC) as necessary for sensory preconditioning (Fournier et al., 2020). However, the role of the RSC for each phase of sensory preconditioning (preconditioning, first-order conditioning and retrieval) is unclear. Additionally, little is known with regards to how of cortical neural activity may facilitate the encoding of neutral stimulus-stimulus associations during the preconditioning. We will use chemogenetics (Aim 1) to determine if the RSC is necessary during each phase of sensory preconditioning. Electrophysiology (Aim 2) will be used to characterize cortical neural activity across six brain regions bilaterally (RSC, dHP, ACC, OFC, PER, and POR) during the preconditioning phase. Additionally, we will explore the impact of extinction following the preconditioning and first-order conditioning phases. This will allow us to determine how extinction influences responding to the higher-order cue during the retrieval test (Aim 3).

Aim 1: Determine if all phases of sensory preconditioning (preconditioning, first-order conditioning, retrieval) require the retrosplenial cortex. Rational: Disruption of the RSC via permeant lesion prior to preconditioning has been shown to abolish sensory preconditioning (Fournier et al., 2020; Robinson et al., 2011). It is unknown if the RSC is required for all phases of unimodal (auditory) appetitive sensory preconditioning. This aim seeks to determine if the RSC is required during each phase through the use of temporary inactivation. Methodology: In three experiments, we will use chemogenetics to inactivate the RSC during each individual phase (preconditioning, first-order conditioning, and test) of an appetitive sensory preconditioning paradigm. Hypothesis: We hypothesize that inactivation of the RSC during preconditioning and retrieval, but not first-order conditioning will disrupt the sensory preconditioning effect.

Aim 2: Characterize cortical neural activity during preconditioning. Rational: Several cortical areas have been implicated in higher-order conditioning, among them the retrosplenial cortex (RSC), as well as the anterior cingulate cortex (ACC), orbitofrontal cortex (OFC), perirhinal cortex (PER), postrhinal cortex (POR), and the hippocampus (HPC) (Fournier et al., 2020; Hart et al., 2022; Hart et al., 2020; Andrew C. Talk et al., 2002; E. M. Taylor-Yeremeeva et al., 2021; Wong et al., 2019). This aim serves to classify the cortical neural activity as the animal learns neutral stimulus-stimulus associations during the preconditioning phase of appetitive sensory preconditioning. Methodology: We will use chronically implanted electrodes to record local field potentials within each brain region bilaterally. We plan to investigate interactions between regions through analysis of coherence for all permutations of recording pairs during stimulus presentation. Hypothesis: We hypothesize that the LFP coherence between regions will change through the preconditioning phase, with activity in early trials being evenly distributed amongst the cortical regions, but in later trials we expect certain regions to become more coherent than others.

Aim 3: Investigation into the effect of extinction of first-order and higher-order cues in sensory preconditioning. Rational: Prior work by Rescorla and Freberg (1978) investigated conditions in which they posited would “destroy” the neutral association formed preconditioning. However, contemporary theory now views extinction as new learning, as opposed to destruction or the original association (Bouton, 2002). Therefore, we decided to revisit the effect of extinction on sensory preconditioning. Methodology: In this aim, we will examine the effect of extinction either prior to or after first-order conditioning. Hypothesis: We hypothesize that extinction of the first-order cue both prior to and following first-order conditioning as well as the higher-order cue prior to first-order conditioning will disrupt sensory preconditioning.

RESEARCH STRATEGY

SIGNIFICANCE

Importance of the Problem to be Addressed.

Understanding how higher-order associations form and influence behavior at both a behavioral and neurobiological level is critical for understanding behavioral processes that may contribute to the formation of maladaptive behaviors. Higher-order associations are indirect in nature making them potentially difficult to identify outside the laboratory. Therefore, they pose a challenge in treating disorders as many treatment strategies rely on identifying triggering stimuli. Here, we have chosen to study higher-order associations through the lens of appetitive sensory preconditioning.

In sensory preconditioning, Cue 1 and Cue 2 are first paired together. Cue 1 is subsequently conditioned via pairings with an unconditioned stimulus. After which Cue 2 is able to elicit behavioral responding despite never having been directly paired with the unconditioned stimulus. This could be particularly important in binge eating, food addiction or substance use disorders as treatments require individuals to identify environmental stimuli that may trigger cravings in order to construct strategies to decrease the likelihood of relapse (Adams et al., 2019; McHugh et al., 2010). If an individual is unable to identify stimuli that have formed higher-order associations, these hidden associations may continue to elicit cravings that undermine treatment efforts. A comprehensive understanding of higher-order associations from both behavioral and neuroscientific perspectives will enable the development of more efficacious therapeutic protocols by allowing us to target associative memories that may remain hidden in various disorders.

Recent research has demonstrated that the retrosplenial cortex (RSC) is required for sensory preconditioning (Fournier et al., 2020; Robinson et al., 2011). These initial studies used pre-training permanent lesions, and therefore the RSC was damaged for the entire behavioral experiment. However, little is known about the involvement of the RSC during each unique *phase* of sensory preconditioning (preconditioning, first-order conditioning, and retrieval). Furthermore, there is limited understanding of the cortical neural activity that supports the encoding of neutral stimulus-stimulus associations. The aims proposed in this study are threefold: to characterize the role of the RSC across all phases of sensory preconditioning, to characterize the neural activity of several cortical regions as neutral stimulus-stimulus associations are learned, and to explore how extinction impacts responding to higher-order cues. This will be accomplished by utilizing chemogenetics (Aim 1), electrophysiology (Aim 2), and behavioral manipulations (Aim 3).

We anticipate that the RSC is required for the preconditioning and retrieval but not the first-order conditioning phase of our sensory preconditioning paradigm (Figure. 1). Additionally, we expect to see change in the coherence between cortical regions across sessions of preconditioning. This would suggest that specific brain regions support encoding of neutral stimulus-stimulus associations. As for extinction of either the preconditioned or first-order conditioned cue, we hypothesize that extinction of the first-order or higher-order cue prior to first-order conditioning, and extinction of the first-order cue following first-order conditioning will disrupt sensory preconditioning.

Significance of the Expected Research Contribution.

We will expand knowledge of the neuronal and behavioral basis of sensory preconditioning. Further, this will help us understand the nature of higher-order associations, laying the foundation for the future development of treatments for disorders that involve higher-order associative processes.

INNOVATION

While appetitive conditioning is adaptive, it also plays a role in the formation of maladaptive behaviors. One example of this is the inflation of reward-seeking behavior that arises through Pavlovian conditioning in which hyper palatable foods or drugs function as strong reinforcers (Itzhak et al., 2014; Lynch et al., 1973; Schulte et al., 2016). Therapies designed to treat such disorders rely on one's ability to identify triggering behaviors and stimuli. This allows for the construction of behavioral strategies to overcome the retrieval of detrimental memories that may elicit things like food and drug craving (Adams et al., 2019; McHugh et al., 2010). The proposed research will investigate the nature of higher-order

memories as well as the neural mechanisms that support them. This knowledge may lead to new therapeutic strategies for disorders that involve higher-order associative processes.

This work utilizes an established appetitive sensory preconditioning paradigm in tandem with chemogenetic and electrophysiological techniques. This combination and the use of novel tools for simultaneous recording of local field potentials from multiple cortical regions, is the basis for a study of the mechanisms of sensory preconditioning that is the first of its kind. We will be manufacturing custom multisite electrodes which will be used in conjunction with established signal processing framework to analyze real-time event encoded electrophysiological measurements of LFP activity.

APPROACH

Aim 1: Determine if all phases of sensory preconditioning (preconditioning, first-order conditioning, retrieval) require the retrosplenial cortex.

Background

Prior research has established that disruption of the RSC through pre-training lesions abolishes sensory preconditioning (Fournier et al., 2020; Robinson et al., 2011). Additionally, later work by Robinson et al. (2014) utilized chemogenetics to temporarily inactivate the RSC at the time of preconditioning which disrupted sensory preconditioning. This has not been extended to the other phases of sensory preconditioning i.e., during first-order conditioning or retrieval. Therefore, we plan to investigate the RSC's role in sensory preconditioning by temporarily inactivating it during each phase of the sensory preconditioning paradigm (preconditioning, first-order conditioning, and retrieval) in 3 separate experiments.

Experiment 1.1 Inactivation during retrieval.

The retrieval phase of our sensory preconditioning protocol is comprised of repeated non reinforced trials of cue A and C. Greater responding to cue A than C would indicate sensory preconditioning as A was paired with the reinforced first-order cue. To explore the RSC's role in retrieval, we have chosen to temporarily inactivate the RSC during this phase. We expect this to disrupt the sensory preconditioning effect, as the RSC has been implicated in memory retrieval (Corcoran et al., 2011).

Experiment 1.2 Inactivation during first-order conditioning.

The conditioning phase of our sensory preconditioning design involves repeated presentations of Cue B paired with a US (sucrose pellet) and cue D unpaired. It is unknown if inactivation during this phase will impact higher-order responding. We will investigate this through temporary inactivation of the RSC during this phase only. We expect that this will not disrupt sensory preconditioning as first order conditioning involves discrete pairings of cue with an outcome, which temporary inactivation studies have shown that the RSC is not necessary for many forms of delay conditioning (Fournier et al., 2021).

Experiment 1.3 Inactivation during preconditioning.

The preconditioning phase consists of repeated pairings of two neutral stimuli presented in sequence such that cues B and D directly follow the termination of cues A and C respectively. Temporary inactivation of the RSC at the time of preconditioning has been shown to disrupt sensory preconditioning, however, it was performed in a sequential bimodal behavioral paradigm in which there was a light and a sound cue (Robinson. et al., 2014). Experiment 3 will examine the necessity of the RSC during this phase, using the same behavioral procedures as Experiments 1.1 and 1.2. This will allow for a more direct comparison of results. We expect this manipulation to disrupt sensory preconditioning as the RSC has been shown to encode other non-cue-outcome information such as spatial representations (Czajkowski et al., 2014).

Overall, we predict *that inactivation of the RSC during the preconditioning and retrieval phases but not the first-order conditioning phase of sensory preconditioning independently will disrupt the sensory preconditioning effect.* Our experimental approach will utilize chemogenetics to temporarily inactive the RSC during each discrete behavioral phase in separate experiments. This aim will provide insight into the neural mechanism that supports higher-order memories of the appetitive nature. We expect that upon the completion of Aim 1, we will have identified the RSC's role in each phase of the

sensory preconditioning paradigm. Understanding the neural substrates of higher-order memories may inform the link between higher-order associations and maladaptive behaviors.

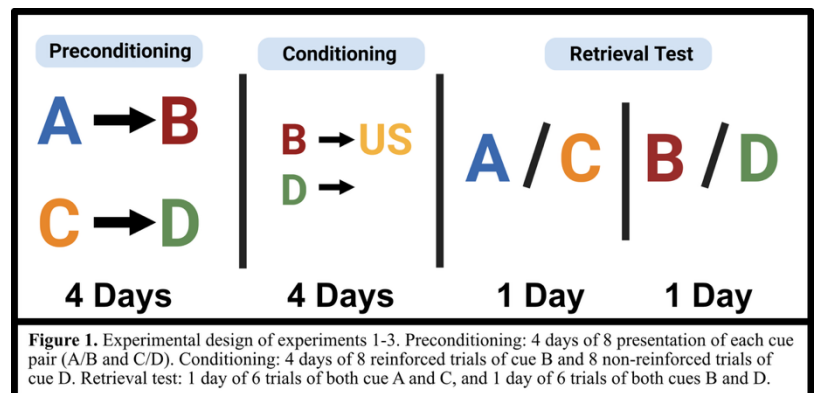
Design

We will utilize Chemogenetics (Designer Receptors Exclusively Activated by Designer Drugs; DREADDs) to temporarily silence the RSC during the discrete phases of our sensory preconditioning behavioral paradigm. For each of the three experiments, two groups of rats will receive viral infusions. The experimental group RSC-hM4Di, will receive 4 bilateral infusions (8 total) of a viral construct (pAAV8-hSyn-hM4Di-mCherry; abbreviated hM4Di) which contains DNA that encodes for an inhibitory DREADDs receptor. The control group RSC-mCherry, will have the same pattern on infusions but with a control virus (pAAV8-hSyn-mCherry; abbreviated mCherry) which does not encode for the inhibitory receptor. Following 3-4 weeks of recovery the rats will receive 2 weeks of a restrictive food taper resulting in maintenance of 85% of the rats free feed body weight. Subsequently the rats will undergo behavioral training (See Figure 1). During the “preconditioning phase” the rats will be presented with cue A immediately followed by cue B ($A \rightarrow B$) and C followed by D ($C \rightarrow D$). The cues will be 10s auditory cues of either; high tone, low tone, white noise, or click. For half the rats, A/B were high tone and low tone, C/D were click and noise; for the other half of the rats this will be reversed such that A/B were click and noise, C/D were high-tone and low-tone. The cues will be counterbalanced within each pair so that each cue type was equally represented as A, B, C or D. During the conditioning phase, B will be terminated with the delivery of a sucrose pellet, and D will be presented alone. B will be directly conditioned, and D will act as the unpaired control. The retrieval phase will consist of 2 days. On the first day the rats will receive trials of A and C and on the second they will receive trials of B and D. Successful sensory preconditioning will result in greater responding to A than C. The second day of testing will allow us to determine if inactivation of the RSC disrupted retrieval of the first-order cues. Rats will receive an injection (i.p.) of the hM4Di agonist Deschloroclozapine (DCZ) 30 minutes prior to preconditioning (Exp 1.3), conditioning (Exp 1.2), and retrieval sessions (Exp 1.1).

The within-subject design we will utilize was constructed with the controls required to confirm the presence of sensory preconditioning (see Rizley & Rescorla, 1972). The controls ensure that responding more to A than C at test is due to higher-order conditioning and not generalization across cues. Due to cue counterbalancing, generalization for B would be expected to equally affect A and C. Additionally, responding more to A than C would be attributed to the conditioning of B because non-reinforced presentations of D should not result in greater responding to C. All experiments will use food cup entries as our behavioral measure; entries will be recorded by way of photobeam in the food cup and are defined as a momentary break in the photobeam. Following each experiment all rats will be perfused, their brains cryo-sectioned and analyzed for expression of the fluorescent protein mCherry via fluorescence microscopy.

Analysis

All experiments in Aim 1 will use head entries into the food cup as the measure of responding. The food cup entries will be recorded through the use of a photobeam located in the food cup. A head entry is defined as a momentary break in the photobeam. We will utilize this measure to analyze the animal’s behavior for the presence of the sensory preconditioning phenomenon. To determine that the behavior was conducted successfully we will be analyzing each phase of the experiment. For the preconditioning phase, we will utilize an elevation ratio in which the responding to CS (10s) is compared to the Pre-CS (10s) period to create a measure of increase in responding from Pre-CS. The elevation ratio will be calculated with the following formula: $(1 + CS) / (2 + (Pre-CS) + CS)$. We will run a repeated measures ANOVA on the elevation ratio collapsed within days between the hM4Di and control groups. The preconditioning phase is void of reinforcer so this analysis will



allow us to make sure that there is no responding that develops over this phase for either group nor cue. For conditioning, we will construct an elevation ratio between the CS (10s) presentation and the Pre-CS period (10s). The formula for the ratio is as follows: $(1 + \text{CS}) / (2 + (\text{Pre-CS}) + \text{CS})$. We will perform a repeated measures ANOVA comparing cues B and D between each group. Since cue B is reinforced with a sucrose pellet, this analysis will allow us to determine if our conditioning was successful by way of significantly more responding to B than D. We will also use this analysis to determine if the two groups have conditioned equivalently. Finally, in order to detect sensory preconditioning in the retrieval test, we will be analyzing both the CS and post-CS periods of the cue presentation as the 10 seconds following the presentation of the preconditioned cue corresponds to when the animal would expect the presentation of the first-order cue. We have chosen to look at the post-CS period because the behavioral activity may be coordinated temporally to when it expects the pellet to occur. Work on the anterior cingulate cortex has demonstrated that neural activity to cue A peaked when the pellets would have been delivered with respect to cue B and its initial neutral pairing with A (Hart et al., 2022). We will construct an elevation ratio in which both the CS (10s) and Post-CS (10s) periods are utilized and compared to the Pre-CS period (20s). The formula for the ratio is as follows: $(1 + \text{CS} + \text{Post-CS}) / (2 + (\text{Pre-CS}) + \text{CS} + \text{Post-CS})$. This ratio renders our measure agnostic to when the rats are responding either during or after the CS. We will utilize a 2 Group (hM4Di vs control) $\times$ 2 Cue (A vs C) $\times$ 6 Trial repeated measures ANOVA, to detect the presence of the sensory preconditioning phenomenon which would manifest in higher responding to A than C. Additionally, this analysis will allow us to determine if the inactivation of the RSC had any effect on sensory preconditioning. Histological preparations of the brain tissue will be analyzed to determine if viral expression was accurately confined to the RSC bilaterally and robust.

Expected Outcomes

Our hypothesis is that the RSC is required for the preconditioning and retrieval phases but not the first-order conditioning phase of sensory preconditioning. We expect chemogenetic inactivation of the RSC during preconditioning and retrieval to disrupt sensory preconditioning resulting in a nonsignificant difference in responding between cue A and C at the time of retrieval test.

Experiment 1.1 Inactivation during retrieval.

During final retrieval we predict that inactivation of the RSC will disrupt the sensory preconditioning effect. This prediction is based on prior research demonstrating importance of RSC in the retrieval of other forms of memory (Corcoran et al., 2011). We predict that inactivation of the RSC will not impact retrieval of the first-order cues.

Experiment 1.2 Inactivation during first-order conditioning.

We expect that inactivation of the RSC at the time of conditioning will not disrupt sensory preconditioning during retrieval. Previous work has demonstrated that the RSC is not necessary for encoding discrete cue-outcome associations (Fournier et al., 2021).

Experiment 1.3 Inactivation during preconditioning.

We predict that inactivation of the RSC at the time of preconditioning will disrupt sensory preconditioning. This would manifest as a non-significant difference in responding to cue A than to cue C.

Alternative Outcomes and Follow-Up Studies

Our hypothesis for this aim emphasizes the importance of the RSC in sensory preconditioning. We predict that the RSC is required for the encoding and retrieval of the stimulus-stimulus association but not the first-order conditioning phase of the sensory preconditioning behavioral paradigm. A possible alternative outcome would be that of no disruption of sensory preconditioning from the inactivation of the RSC during one or multiple of the phases. If we saw no disruption when inactivating, it is possible that different brain regions or memory encoding strategies are being utilized. This will be followed up with Aim 2 in which we record from several brain regions during the preconditioning phase. In order to look at the potential of different brain regions functioning to encode and retrieve these memories we could extend that study to encompass each phase of the sensory preconditioning protocol.

Potential Problems and Alternative Strategies

One possible problem would be that of DREADDs expression within the RSC. This could take the form of inadequate or off target expression. We do not believe this will be an issue based on previously establishing accurate expression and inactivation within the RSC. The RSC is a large cortical region and we have had very high success in targeting it in the lab (Fournier et al., 2021). We follow up every DREADDs study with an in-depth analysis of the histology through fluorescence microscopy, removing animals that had low expression, unilateral expression, or high expression in other brain regions. Another potential problem would be that of the concern of metabolism of our receptor agonist into the biologically active clozapine. In our experiments we make direct comparison to animals that have received a control virus that does not express hM4Di but do get DCZ injections. We have chosen the agonist DCZ due to its higher binding affinity, as it allows us to utilize a dose 20-fold lower than that of clozapine-N-oxide.

Aim 2: Characterize cortical neural activity during the preconditioning phase of sensory preconditioning.

Background

Several cortical regions in addition to the RSC have been implicated in higher order conditioning; the dorsal hippocampus (dHPC), anterior cingulate cortex (ACC), orbitofrontal cortex (OFC), perirhinal cortex (PER), and the postrhinal cortex (POR) (Table 1). To our knowledge no one has investigated the neural activity of these regions and the potential cross-cortical network that may support the encoding of neutral stimulus-stimulus associations during preconditioning. The focus of Aim 2 is to classify this neural activity. We hypothesize that the local field potential (LFP) coherence between and within regions will change through the preconditioning phase of sensory preconditioning. With coherence in early trials being evenly distributed amongst the cortical regions, but in later trials we expect certain regions to become more coherent than others and will show directionality of phase propagation. To investigate this, we will utilize multi-site LFP probes to record LFP activity of all cortical regions mentioned above as well as within brain regions bilaterally in real time. Analysis of this LFP activity will allow us to determine the nature of possible cross-cortical networks with respect to preconditioning, and the relationship between LFPs neutral stimulus-stimulus associations. We expect that upon completion of Aim 2 we will have greater understanding of how cortical regions support higher-order memories through the encoding of neutral associations.

Table 1

Current Knowledge of Cortical Involvement in Preconditioning

<i>Brain Region</i>	<i>Activity During Preconditioning</i>	<i>Necessary for Preconditioning</i>
<i>ACC</i>	Yes	Unknown
<i>OFC</i>	Yes	Yes
<i>dHPC</i>	Yes	Yes
<i>RSC</i>	Unknown	Yes
<i>POR</i>	Unknown	Yes
<i>PER</i>	Unknown	Yes

Note. This table summarizes rodent studies examining the cortical substrates of the preconditioning phase of sensory preconditioning. The “Activity During Preconditioning” column consists of studies reporting neural correlates recorded during this preconditioning phase (Hart et al., 2022; Pinho et al., 2025; Sadacca et al., 2018). The “Necessary for Preconditioning” column refers to studies in which a causal manipulation disrupted each given region demonstrating its necessity for sensory preconditioning (Hart et al., 2020; Nicholson & Freeman, 2000; Robinson et al., 2014; A. C. Talk et al., 2002; Elisa M. Taylor-Yeremeeva et al., 2021). Only rodent studies were included. ACC = anterior cingulate cortex; OFC = orbitofrontal cortex; dHPC = dorsal hippocampus; RSC = retrosplenial cortex; POR = postrhinal cortex; PRH = perirhinal cortex.

Design

For EXP 2, we will implant a group of rats with a multi-site electrode array into the RSC, dHPC, ACC, OFC, PER and POR bilaterally. Following recovery, the rats will then be subjected to the same appetitive sensory preconditioning protocol as Aim 1 (Figure 2).

Neural Recordings and Analysis

We will conduct the neural recordings in two operant chambers. Wide-band neural data (0.1-7500Hz) will be recorded via a Neuralynx Digital Lynx SX system during each session. The LFP Data will be analyzed for amplitude and frequency of oscillations at theta (5-12Hz) and Gamma frequencies (30-90hZ) in each region via time-dependent Fourier transform analysis (MATLAB; Spectrogram). We will analyze coherence between regions in a matrix for all permutations of cortical regions for every stimulus epoch and behavioral response following the methods of prior work (Dickson et al., 2022). This measure will

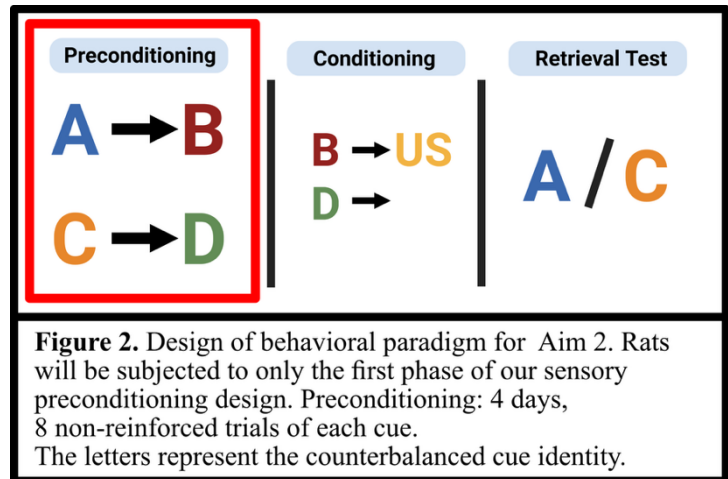
allow us to analyze if each region pair exhibits LFP coherence by phase which has been hypothesized to represent the functional connectivity of neural circuits. We will analyze any potential phase lag of frequency and time between each region allowing us to determine the direction of information flow or influence of one brain region on another (Sauseng & Klimesch, 2008). We will perform within-structure coherence analysis across bilateral electrodes. Additionally we will analyze within channel, examining the amplitude relative to delta, theta and gamma (slow and medium) and cross frequency coupling (Buzsáki & Schomberg., 2015). In order to remove the possible influence of volume conductance with respect to the calculation of coherence we will conduct phase-lag index tests between pairs of each region, we will remove any pairs with a phase index of 0 (Stam et al., 2007). We will compare effects via repeated measures general linear model (GLM) approach, to test for interactions with stimulus presentations and behavioral responses (i.e., Tone x Mag entries). In order to determine that the behavior was conducted successfully we will utilize the same strategy as Aim 1, using the same elevation ratios. In order to determine that the behavior was conducted successfully we will utilize the same strategy as Aim 1, using the same elevation ratios. We will run a repeated measures ANOVA on the elevation scores collapsed within days. The preconditioning phase is void of reinforcer so this analysis will allow us to make sure that there is no responding that develops over this phase. Histological preparations of the brain tissue will be analyzed to determine correct placement of the electrodes.

Expected Outcomes

We predict that over the course of the preconditioning session days, coherence between brain regions will change. Such that LFP activity relative to stimuli will be similar across brain regions, but as training progresses the brain regions will show differential activity and coherence between some but not all brain regions. This would suggest that over the course of training specific brain regions become engaged to support learning of neutral stimulus-stimulus associations.

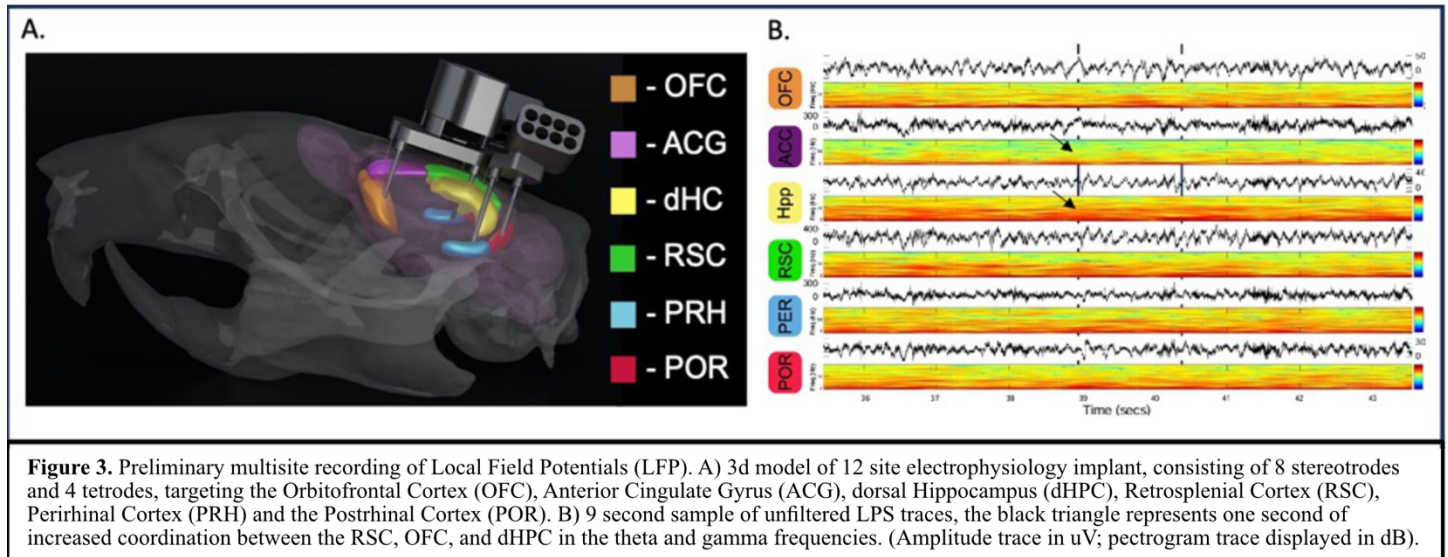
Alternative Outcomes and Follow-Up Studies

A possible alternative outcome is that we observe equal phase coherence during all phases of this experiment. This would suggest that all areas are engaged throughout all phases of sensory preconditioning and are some of many regions of specialized function that work independently. This could suggest that analysis of LFP data might be missing cellular level processing. If this is the case, we will follow up this study utilizing high-density silicon probes implanted into fewer cortical regions with the same behavioral protocol. This would allow us to record single unit activity across a larger area within the targeted brain regions. We would analyze for differential temporal and rate coding amongst neurons across the cortical regions.



Potential Problems and Alternative Strategies

A potential problem of this study would be that of coherence being due to the behavior itself either postural attention or movement. In order to rule this out we could compare the LFP coherence between brain regions during cue presentations with activity that occurred during periods of movement in the inter-trial intervals. If increased coherence was due to the behavior itself, we would expect to see equivalent changes during both cue presentations and the non-cued movement. Another potential problem is technical feasibility. Our lab has conducted pilot research in the design, manufacture and implantation of a multisite electrophysiological implant. We were able to target the six brain regions bilaterally and acquire LFP data in which we were able to detect neural coordination between the RSC, orbital frontal cortex and the dorsal hippocampus (Figure 3).



Aim 3: Investigation into the effect of extinction of first-order and higher-order cues in sensory preconditioning.

Background

Rescorla and Freberg (1978), investigated what they posited as conditions that would “destroy” the preconditioned associations in sensory preconditioning. They found that pre-extinction of either cue prior to first-order conditioning disrupted sensory preconditioning leading them to the conclusion that they disrupted the association. However, we know that extinction is not unlearning. This us to reexamine the effect of extinction on sensory preconditioning. The focus of this aim is to determine if extinction of the first-order cue (Exp 3.1 & 3.3) prior too or following the first-order conditioning phase and extinction of the higher-order cue (Exp 3.2) prior to the first-order conditioning phase disrupts the sensory preconditioning effect (Figure 4). Experiments 3.1 and 3.2 consist of pre-extinguishing cues A and B prior to first-order conditioning. This allows us to test the effect extinction has on sensory preconditioning with respect to the neutral A-B association. We hypothesize that sensory preconditioning requires this association between A-B therefore we expect extinction of either of these cues prior to first order conditioning will disrupt the sensory preconditioning effect. Experiment 3.3 was devised to investigate the effect of extinction has on sensory preconditioning through the extinction of the B-US association. This has been investigated in conditioned taste aversion in rats, however, to our knowledge these effects have not been extended to appetitive sensory preconditioning. We hypothesize that extinction of B following first-order conditioning will disrupt sensory preconditioning.

In order to investigate this, we will employ the same behavioral paradigm listed previously with the addition of a cue pre-extinction session (preceding first-order conditioning) or extinction session (following first-order conditioning). This will be conducted in 3 separate experiments, Exp 3.1; pre-extinction of A preceding first-order conditioning., Exp 3.2; pre-extinction of B preceding first-order conditioning, Exp 3.3; extinction of B following first-order conditioning (Figure 4). We expect that upon the completion of Aim 3, we will have determined if extinction of the neutral stimulus-stimulus

association between cue A and cue B disrupts sensory preconditioning and if disrupting the excitatory association between cue B and the US disrupts sensory preconditioning as well.

Design

For experiment 3.1-3 we will be using separate groups of naive rats in each experiment. We will be using the same behavioral paradigm used in the experiments with the addition of another phase, either pre-extinction (Exp 3.1 and 3.2) or extinction (Exp 3.3) see Figure 4. For experiments 3.1 and 3.2 following preconditioning, half of the rats will receive 2 days of 16 repeated non reinforced presentations of cue A or B alone respectively. The other half of the rats will receive a time matched session (90 minutes) in the context without any cue presentations. Experiment 3.3 will follow a similar design to experiments 3.1 and 3.2 except that the extinction session will be following first order conditioning.

Analysis

Analysis of behavioral responding will follow Aim one with slight alteration. Our two groups will be extinction and context exposure. Additionally, there will be the addition of an extinction session either preceding (Exp 3.1, 3.2) or succeeding (Exp 3.3) the conditioning session. For the analysis of the extinction sessions, we will utilize an elevation ratio between the Pre-CS period (20s) and the CS (10s) presentation, for the control group we will record activity during the equivalent 10s periods that would correspond to the cue presentations in the extinction group.

Expected Outcomes

Our hypothesis is twofold, pre-extinction of cues A and B both prior to first-order conditioning will disrupt sensory preconditioning and, we hypothesize that extinction of the higher-order cue B following first-order conditioning will also disrupt sensory preconditioning.

1. Pre-extinction of A or B prior to conditioning (Exp 3.1 & 3.2)

In these two experiments we will test the effect of extinguishing the neutral stimulus-stimulus association by pre-extinguishing either cue A or B prior to first-order conditioning. We expect that extinction of this neutral stimulus-stimulus association between the cues will disrupt sensory preconditioning. Behaviorally this would result in a non-significant difference in responding to cues A and C in the retrieval test.

2. Extinction of B following first-order conditioning (Exp 3.3)

In this last experiment we will examine the effect of extinguishing the excitatory association between cue B and the US. We will test this by extinguishing cue B following first-order conditioning and examine its effect on sensory preconditioning in the retrieval test. We expect that this manipulation will disrupt sensory preconditioning. As with the previous two experiments, this would manifest in a non-significant difference in responding to cues A or C at test. This would suggest that for sensory preconditioning to occur the excitatory association between B and the US must be intact for sensory preconditioning to occur.

Alternative Outcomes and Follow-Up Studies

One possible alternative outcome is that the pre-extinction of either cue doesn't disrupt sensory preconditioning. This would suggest that repeated exposure of either cue prior to conditioning doesn't affect the associative process occurring during preconditioning that drives the sensory preconditioning effect. A logical follow up study would be that of extinguishing A following the first order conditioning phase. This would require the addition of a test phase in which responding to the first-order cue B is compared between a group that received extinction of A and one that received time matched context exposure in an extinction phase following conditioning. If there was a disruption of the sensory preconditioning in this follow-up, it would suggest that the stimulus-stimulus association is required for sensory preconditioning to occur but only after some associative process takes place during first order conditioning.

Potential Problems and Alternative Strategies

We do not foresee any issues with feasibility as we have successfully performed several sensory preconditioning experiments with the general protocol previously. However, one potential problem could be in our behavioral preparation for experiment 3.2. Latent inhibition is the phenomenon in which repeated nonreinforced pre-exposure to a CS results in a decrement in the learning of an association with that CS and a US (Lubow & Moore, 1959). In our case, extinction of B

prior first-order conditioning (Exp 3.2.) may cause a decrement to learning the B-US association through latent inhibition. We will monitor response to B during first order conditioning in both groups to ensure that they both reach the same level of conditioned responding.

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